56. Drug Dosing Considerations in Patients With Acute Kidney Injury and Chronic Kidney Disease · assets 1-9 / 18

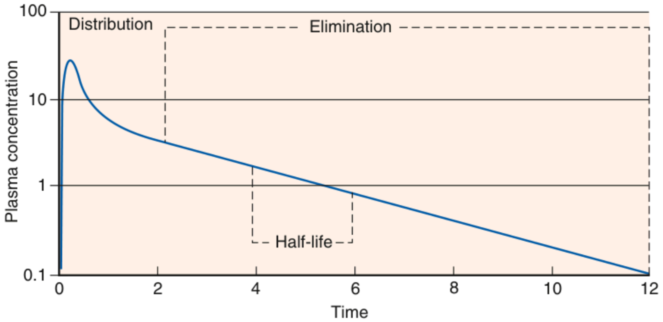

Fig. 56.1 Distribution and elimination of a drug after intravenous administration.

Fig_56_1

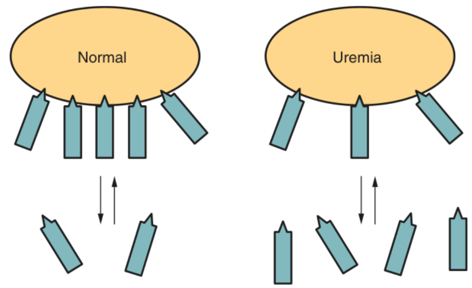

Fig. 56.2 Decreased protein binding in uremia. Displacement of the drug from its binding site by an accumulation of undefined uremic toxins or a uremia-induced conformational change in the binding-site geometry results in more free drugs in the plasma.

Fig_56_2

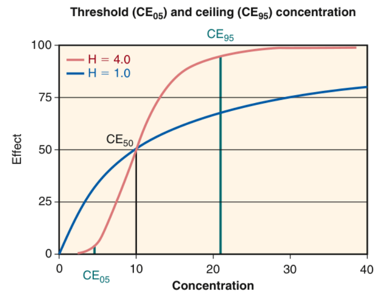

Fig. 56.3 Threshold concentration, *CE_{05}*, producing 5% of the maximum effect, and ceiling concentration, *CE_{95}*, producing 95% of the maximum effect. With a Hill coefficient of H = 1.0, CE_{05} = 0.5 and CE_{95} = 190, whereas for H = 4.0, the threshold is higher, with CE_{05} = 6.0, but the ceiling is much less, with CE_{95} = 21 mg/L.

Fig_56_3

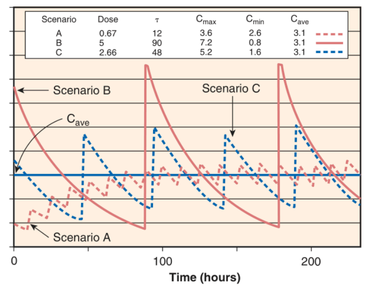

Fig. 56.4 Although the average steady-state concentrations *(C_{ave})* are identical regardless of which dosage adjustment strategy one decides to use, the concentration–time profile will be markedly different if one changes the dose and maintains the dosing interval *(τ)* constant *(Scenario A)*, versus changing the dosing interval and maintaining the dose constant *(Scenario B)*, or changing both *(Scenario C)*.

Fig_56_4

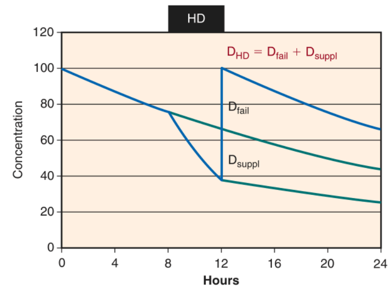

Fig. 56.5 Supplementary dose after hemodialysis *(HD)*. To maintain therapeutic target concentrations, a supplementary dose must be given after hemodialysis to replace the removed fraction of the dose. The dose after dialysis *(D_{HD})* combines both the adjusted maintenance dose *(D_{fail})* and supplementary dose *(D_{suppl})*.

Fig_56_5

Table 56.1 Volume of Distribution of Selected Drugs in Patients With Normal Kidney Function and Those on Dialysis

Drug	Normal (L/kg)	Stage 5 Chronic Kidney Disease (L/kg)	Change From Normal (%)
Increased			
Amikacin	0.20	0.29	45
Cefazolin	0.13	0.17	31
Cefoxitin	0.16	0.26	63
Ceftriaxone	0.28	0.48	71
Cefuroxime	0.20	0.26	30
Dicloxacillin	0.08	0.18	125
Erythromycin	0.57	1.09	91
Furosemide	0.11	0.18	64
Gentamicin	0.20	0.32	60
Isoniazid	0.6	0.8	33
Minoxidil	2.6	4.9	88
Naproxen	0.12	0.17	42
Phenytoin	0.64	1.4	119
Trimethoprim	1.36	1.83	35
Vancomycin	0.64	0.85	33
Decreased			
Atenolol	1.2	0.9	–25
Chloramphenicol	0.87	0.60	–31
Ciprofloxacin	2.5	1.95	–22
Digoxin	7.3	4.0	–45
Ethambutol	3.7	1.6	–57
Methicillin	0.45	0.3	–33
Metoprolol	5.6	1.0	–82
Pindolol	2.1	1.1	–48
Propranolol	4.4	3.6	–18

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bomback A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy* 2009;29:562–577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; Verbeeck RK, Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65:757–773; and Olyaei AJ, Steffl JL. A quantitative approach to drug dosing in chronic kidney disease. *Blood Purif*. 2011;31:138–145.

Table_56_1

Table 56.2 Unbound Fraction of Selected Drugs in Patients With Normal Kidney Function and End-Stage Kidney Disease (ESKD)

Drug	Normal Patient	ESKD Patient	Change From Normal (%)
Acidic Drugs			
Abecarnil	4	15	275
Azlocillin	62.5	75	20
Cefazolin	16	29	81
Cefoxitin	27	59	119
Ceftriaxone	10	20	100
Clofibrate	3	9	200
Dicloxacillin	3	9	200
Diflunisal	12	44	267
Doxycycline	12	28	133
Furosemide	4	6	50
Methotrexate	57.2	63.8	12
Metolazone	5	10	100
Moxalactam	48	64	33
Pentobarbital	34	41	21
Phenytoin	10	21.5	115
Salicylate	8	20	150
Sulfamethoxazole	34	58	71
Valproic acid	8	23	188
Warfarin	1	2	100
Basic Drugs			
Decreased			
Bepridil	0.3	0.1	–67
Clonidine	55.6	47.6	–14
Disopyramide	32	28	–13
Propafenone	3.4	2.4	–29
Increased			
Amphotericin B	3.5	4.1	17
Chloramphenicol	45	64	42
Clonazepam	13.9	16	15
Diazepam	2	8	300
Fluoxetine	5.5	6.5	18
Ketoconazole	1	1.5	50
Prazosin	6	10.1	68
Rosiglitazone	0.16	0.22	38
Triamterene	19	43	126

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bomback A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy* 2009;29:562–577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; and Verbeeck RK, Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65:757–773.

Table_56_2

Table 56.3 Drugs With Pharmacologically Active Metabolites That May Affect Efficacy or Toxicity in Patients With Severe Chronic Kidney Disease

Parent Drug	Metabolite	Pharmacologic Activity of Metabolites
Acetaminophen	*N*-Acetyl-*p*-benzoquinoneimine	Responsible for hepatotoxicity
Allopurinol	Oxipurinol	Metabolite primarily responsible for suppression of xanthine oxidase
Azathioprine	Mercaptopurine	All immunosuppressive activity resides in the metabolite
Cefotaxime	Desacetyl cefotaxime	Similar antimicrobial spectrum, but 10% to 25% as potent
Chlorpropamide	2-Hydroxychlorpropamide	Similar in vitro insulin-releasing activity
Clofibrate	Chlorophenoxyisobutyric acid	Primarily responsible for hypolipidemic effect and direct muscle toxicity
Codeine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in kidney failure patients
Imipramine	Desmethylimipramine	Similar antidepressant activity
Ketoprofen	Ketoprofen glucuronide	Accumulation of acyl glucuronide may worsen toxic effects (GI disturbances, impairment of kidney function)
Meperidine	Normeperidine	Less analgesic activity than parent, but more central nervous system stimulatory effects, epileptogenic
Morphine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in ESKD
Mycophenolic acid	Mycophenolic acid glucuronide	Lacks pharmacologic activity but may be associated with dose-limiting (GI) side effects
Procainamide	*N*-Acetyl procainamide	Distinct antiarrhythmic activity; mechanism different from that of parent compound
Sulfonamides	Acetylated metabolites	Devoid of antibacterial activity; elevated concentrations associated with increased toxicity
Theophylline	1,3-Dimethyl uric acid	Cardiotoxicity has been demonstrated
Zidovudine	Zidovudine triphosphate	Primarily responsible for antiretroviral activity

ESKD, End-stage kidney disease; *GI,* gastrointestinal.

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bomback A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; and Battistella M, Nolin TD. Drug Therapy Individualization for Patients with Chronic Kidney Disease. In: Joseph T. DiPiro, Gary C. Yee, L. Michael Posey, et al, eds. *Pharmacotherapy: A Pathophysiologic Approach*. 11th ed. New York: McGraw-Hill; 2020.

Table_56_3

Table 56.4 Major Pathways of Nonkidney Drug Clearance (Cl_{NR})

Cl_{NR} pathway	Selected Substrates
Oxidative Enzymes	
CYP1A2	Polycyclic aromatic hydrocarbons, caffeine, imipramine, theophylline
CYP2A6	Coumarin
CYP2B6	Nicotine, bupropion
CYP2C8	Retinoids, paclitaxel, repaglinide
CYP2C9	Celecoxib, diclofenac, flurbiprofen, indomethacin, ibuprofen, losartan, phenytoin, tolbutamide, *S*-warfarin
CYP2C19	Diazepam, *S*-mephenytoin, omeprazole
CYP2D6	Codeine, debrisoquine, desipramine, dextromethorphan, fluoxetine, paroxetine, duloxetine, nortriptyline, haloperidol, metoprolol, propranolol
CYP2E1	Ethanol, acetaminophen, chlorzoxazone, nitrosamines
CYP3A4/5	Alprazolam, midazolam, cyclosporine, tacrolimus, nifedipine, felodipine, diltiazem, verapamil, fluconazole, ketoconazole, itraconazole, erythromycin, lovastatin, simvastatin, cisapride, terfenadine
Conjugative Enzymes	
UGT	Acetaminophen, morphine, lorazepam, oxazepam, naproxen, ketoprofen, irinotecan, bilirubin
NAT	Dapsone, hydralazine, isoniazid, procainamide
Transporters	
OATP1A2	Bile salts, statins, fexofenadine, methotrexate, digoxin, levofloxacin
OATP1B1	Bile salts, statins, fexofenadine, repaglinide, valsartan, olmesartan, irinotecan, bosentan
OATP1B3	Bile salts, statins, fexofenadine, telmisartan, valsartan, olmesartan, digoxin
OATP2B1	Statins, fexofenadine, glyburide
PGP	Digoxin, fexofenadine, loperamide, irinotecan, doxorubicin, vinblastine, paclitaxel, erythromycin
MRP2	Methotrexate, etoposide, mitoxantrone, valsartan, olmesartan
MRP3	Methotrexate, fexofenadine

Data from Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; Yeung CK, Shen DD, Thummel KE, et al. Effects of chronic kidney disease and uremia on hepatic drug metabolism and transport. *Kidney Int*. 2014;85:522–528; Kagaya H, Niioka T, Saito M, et al. Effect of hepatic drug transporter polymorphisms on the pharmacokinetics of mycophenolic acid in patients with severe renal dysfunction before renal transplantation. *Xenobiotica* 2017;47:916–922; Thomson BK, Nolin TD, Velenosi TJ, et al. Effect of CKD and dialysis modality on exposure to drugs cleared by nonrenal mechanisms. *Am J Kidney Dis*. 2015;65:574–582; Joy MS, Frye RF, Nolin TD, et al. In vivo alterations in drug metabolism and transport pathways in patients with chronic kidney diseases. *Pharmacotherapy* 2014;34:114–122; Lee W, Kim RB. Transporters and renal drug elimination. *Annu Rev Pharmacol Toxicol*. 2004;44:137–166; Masereeuw R, Russel FGM. Therapeutic implications of renal anionic drug transporters. *Pharmacol Ther*. 2010;126:200–216; and Hsueh CH, Yoshida K, Zhao P, et al. Identification and quantitative assessment of uremic solutes as inhibitors of renal organic anion transporters, OAT1 and OAT3. *Mol Pharm*. 2016;13:3130–3140.

Table_56_4

Table 56.5 Equations For Estimation of Creatinine Clearance or Glomerular Filtration Rate in Adults With Stable Renal Function

Table_56_5

Table 56.6 Stepwise Approach to Adjust Drug Dosage Regimens for Patients With Impaired Kidney Function

Step	Process	Assessment
1	Obtain history and relevant demographic and clinical information	Record demographic information, obtain medical history (including history of renal disease), and record current laboratory information (e.g., serum creatinine)
2	Estimate creatinine clearance and/or determine eGFR	Use Cockcroft–Gault equation to estimate creatinine clearance, or calculate creatinine clearance from timed urine collection and/or determine eGFR using 2021 CKD EPIcr or CKD EPIcr-cys
3	Review current medications	Identify drugs for which individualization of the treatment regimen will be necessary
4	Calculate individualized treatment regimen	Determine treatment goals (see text); calculate dosage regimen based on pharmacokinetic characteristics of the drug and patient's renal function.
5	Monitor	Monitor parameters of drug response and toxicity; monitor drug levels if available or applicable.
6	Revise regimen	Adjust regimen based on drug response or change in patient status (including kidney function), as warranted.

Adapted from Battistella M, Nolin TD. Drug Therapy Individualization for Patients with Chronic Kidney Disease. In: Joseph T. DiPiro, Gary C. Yee, L. Michael Posey, et al., eds. *Pharmacotherapy: A Pathophysiologic Approach*. 11th ed. New York: McGraw-Hill; 2020.

Table_56_6

Table 56.7 Drugs Best Administered Before Hemodialysis

Drug Class	Examples	Drug Fraction Removed by One Dialysis Session	Reference
Anticancer	Carboplatin	20% dose post-HD; 84% dose before HD	Chatelut et al.209; Kamata et al.210; Yoshida et al.211; Oguri et al.212
	Cisplatin	85%	Watanabe et al.213
	Oxaliplatin	65%	Katsumata et al.214
	Cyclophosphamide	22% (M % unknown)	Haubitz et al.215
	Ifosfamide	70%–87% (M, 72%–77%)	Carlson et al.216
	Capecitabine	FBAL 50%	Walko and Lindley 217
	Gemcitabine	dFdU 50%	Koolen et al.218
	Methotrexate	(M, 36%)	Garlich and Goldfarb 219
	Cytosine arabinoside	39% (M, 52%–63%)	Radeski et al.220
	Topotecan	50%	Herrington et al.221
	Bleomycin	30%–60%	Kamidono et al.223
	Lenalidomide	30%	João et al.224
	Pemetrexed	30%	Izzedine225
	Tegafur = S-1	60%	Tomiyama et al.226
	Carboplatin	84%	Oguri et al.227
	Fludarabine	2F-Ara-A 25%	Kielstein et al.228
	Fluorouracil = 5-FU infusion	FBAL 60%	Rengelshausen et al.229
	Iodine 131	50%	Fofi et al.230
	Irinotecan	SN-38 50%	Koike et al.231
Aminoglycoside	Gentamicin	75%	Veinstein et al.203
	Tobramycin	80%	Kamel et al.177
Contrast agent	Gadolinium	65%–74%	Rodby222

dFdU, 2′,2′-Difluorodeoxyuridine; *FBAL*, α-fluoro-β-alanine; *HD*, hemodialysis; *M*, metabolite.

Table_56_7

Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury

Drug	Degree of Drug Dose Reduction or Interval Prolongation			Dosage Recommendations for Patients Receiving Kidney Replacement Therapy		
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT
Acebutolol	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Acetaminophen	q4h	q6h	q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Acetazolamide	q6h	q12h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Acetohexamide	Avoid	Avoid	Avoid	Avoid	Avoid	Avoid
Acetohydroxamic acid	100%	100%	Avoid	Unknown	Unknown	Unknown
Acetylsalicylic acid	q4h	q4-6h	Avoid	As normal GFR	As normal GFR	Dose as GFR 10-50
Acrivastine	8 mg q6h	8 mg q8-12h	8 mg q12-24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Acyclovir	5 mg/kg q8h	5 mg/kg q12-24h	2.5-5 mg/kg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Allopurinol	100%	50%	33%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Amantadine	q24h	q48-72h	q7days	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Amikacin[a]	5-6 mg/kg q12h	3-4 mg/kg q24h	2 mg/kg q24–48h	5 mg/kg after HD	15-20 mg/L/day	7.5 mg/kg q24h
Amiloride	100%	50%	Avoid	NA	NA	NA
Amlodipine	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Amoxapine	100%	100%	100%	Unknown	Unknown	Dose as GFR 10-50
Amphotericin	q24h	q24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Amphotericin B	q24h	q24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose for GFR 10-50
Amphotericin B lipid	q24h	q24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ampicillin	250 mg-2 g q4-6h	250 mg-2 g q6h	250 mg–1 g q6h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Apixaban	100%	25-50 mL/min: 100% <25 mL/min: Not recommended	Not recommended	5 mg BID; 2.5 mg BID if age >79 years or body weight <61 kg	Unknown	Unknown
Aripiprazole	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Atenolol	100% q24h	50% q24h	25% q24h	Dose as GFR 25-50	Dose as GFR <10	Dose as GFR 10-50
Auranofin	6 mg q24h	3 mg q24h	Avoid	Avoid	Avoid	Dose as GFR 10-50
Azathioprine	100%	75%–100%	50%–100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Aztreonam	100%	50%	25%	Dose as GFR <10	Dose for GFR <10	Dose as GFR 10-50
Benazepril	100%	50%–75%	25%–50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Bezafibrate	50%–100%	25%–50%	Avoid	200 mg q72h	200 mg q72h	200 mg q24-48h
Bisoprolol	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Bleomycin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Bretylium	100%	25%-50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Bumetanide	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Bupropion	100% q24h	100% q24h	100% q24h	Dose as GFR < 10	Dose as GFR <10	Dose as GFR 10-50
Butorphanol	100%	75%	50%	Unknown	Unknown	Dose as GFR 10-50
Canagliflozin	100%	30-50 mL/min: Not Recommended <30 mL/min: Avoid	Avoid	Avoid	Unknown	Unknown
Capreomycin	q24h	q24h	q48h	Dose as GFR <10	Dose as GFR <10	As normal GFR
Captopril	100% q8-12h	75% q12-18h	50% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

Continued on following page

Table_56_8_part1

Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

Drug	Degree of Drug Dose Reduction or Interval Prolongation			Dosage Recommendations for Patients Receiving Kidney Replacement Therapy		
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT
Carboplatin	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Carteolol	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cefaclor	100%	100%	50%-100%	250-500 mg q8h	250 mg q8-12h	Dose as GFR 10-50
Cefadroxil	q12h	q12h	q24h	0.5-1.0 g after HD	0.5 g/day	Dose as GFR 10-50
Cefamandole	q6h	q6-8h	q8-12h	0.5-1.0 g q12h	0.5-1.0 g q12h	Dose as GFR 10-50
Cefazolin	q8h	q12h	50% q24-48h	15-20 mg/kg after HD	Dose as GFR 10-50	Dose as GFR 10-50
Cefepime	q12h	50%-100% q24h	25%-50% q24h	Dose as GFR <10	Dose for GFR <10	1-2 g q12h
Cefixime	100%	75%-100%	50%	Dose as GFR <10	Dose for GFR <10	Dose as GFR 10-50
Cefotaxime	q6h	q6-12h	1 g q8-12h	Dose as GFR <10	Dose as GFR <10	1-2 g q12h
Cefotetan	q12h	q24h	q48h	1 g after HD	1 g q24h	Dose as GFR 10-50
Cefoxitin	q6–8h	q8-12h	q24-48h	1 g after HD	1 g q24h	Dose as GFR 10-50
Cefpodoxime	100%	100%	100-200 mg q24-48h	Dose as GFR <10	Dose as GFR <10	As normal GFR
Cefprozil	100%	50% q12h	50% q12h	250 mg after HD	Dose as GFR <10	Dose as GFR <10
Ceftazidime	100%	1-2 g q24h	0.5-1 g q48h	1 g after HD	0.5-1 g q24h	1-2 g q12h
Ceftibuten	100%	50%	25%	400 mg after HD	Dose as GFR <10	Dose as GFR 10-50
Ceftizoxime	q8h	q12h	q24h	1 g after HD	0.5-1.0 g q24h	Dose as GFR 10-50
Cefuroxime (IV)	100% q8h	q8-12h	750 mg q12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Celiprolol	100%	100%	75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Cephalexin	250-500 mg q6h	250-500 mg q8-12h	250-500 mg q12-24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cephradine	100%	50%	25%	Dose as GFR <10	Dose for GFR <10	As normal GFR
Cetirizine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Chloroquine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Chlorpropamide	50%	Avoid	Avoid	Avoid	Avoid	Avoid
Chlorthalidone	q24h	Avoid	Avoid	Avoid	Avoid	Unknown
Cibenzoline	100% q12h	100% q12h	66% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cidofovir	50%-100%	Avoid	Avoid	No data	No data	Avoid
Cilazapril	75% q24h	50% q24-48h	10%-25% q72h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cimetidine	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ciprofloxacin	100%	50%-100%	50%	250 mg q12h	250 mg q8h	200 mg IV q12h
Cisplatin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clarithromycin	100%	75%	50%-75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clodronate	100%	50%	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clofazimine	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clofibrate	q6–12h	q12–18h	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clomipramine	100%	Start at lower dose, monitor effect	Start at lower dose, monitor effect	Dose as GFR 10-50	Dose as GFR 10-50	Dose as GFR 10-50
Clonidine	q12h	q12-24h	q24h	As normal GFR	As normal GFR	As normal GFR
Clopidogrel	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Codeine	100%	75%	50%	As normal GFR	As normal GFR	As normal GFR
Colchicine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cyclophosphamide	100%	75%-100%	50%-75%	50%	75%	100%
Cycloserine	q12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

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Dabigatran	100%	15-50 mL/min: 75 mg BID	Not Recommended	Not Recommended	Unknown	Unknown
Dapsone	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Daunorubicin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Didanosine	50%–100%	33%–50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Diflunisal	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Digitoxin	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Digoxina	100% q24h	25%-50% q24h	10%-25% q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Disopyramide	q8h	q12h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Dobutamine	100%	100%	100%	As normal GFR	As normal GFR	As normal GFR
Doxacurium	100%	50%	50%	Unknown	Unknown	Dose as GFR 10-50
Dyphylline	75%	50%	25%	Dose as GFR <10	Unknown	Dose as GFR 10-50
Emtricitabine	q24h	q48-72h	q96h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Enalapril	100%	50%-100%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ertapenem	100%	100%	50%	Dose as GFR <10	Dose as GFR < 10	Dose as GFR 10–50
Erythromycin	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Ethambutol	q24h	q24-36h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ethchlorvynol	100%	Avoid	Avoid	Dose as GFR <10	Dose as GFR <10	NA
Ethionamide	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ethosuximide	100%	100%	75%-100%	As normal GFR	As normal GFR	As normal GFR
Etoposide	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Famciclovir	100%	q12-24h	50% q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Famotidine	100%	50%	20 mg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fentanyl	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fexofenadine	q12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Flecainide	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fluconazole	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Flucytosine	50 mg/kg q12h	50 mg/kg q24h	50 mg/kg q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fludarabine	75%-100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Foscarnet	28 mg/kg/q8h	15 mg/kg/q8h	6 mg/kg/q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fosinopril	100%	100%	75%-100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Gabapentin	300-600 mg q8h	200-700 mg q12h	100-300 mg q24h	LD: 300 mg MD: 100-300 mg q24h	As normal GFR	As normal GFR
Gallamine	75%	Avoid	Avoid	NA	NA	Avoid
Ganciclovir	2.5-5 mg/kg q12h	1.25-2.5 mg/kg q24h	1.25 mg/kg q24h	Dose as GFR <10	Dose as GFR <10	2.5 mg/kg q24h
Gemfibrozil	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Gentamicina	5-7 mg/kg/day	2-3 mg/kg/day by levels	2 mg/kg q48-72h by levels	3 mg/kg after HD by levels	3-4 mg/L/day by levels	Dose as GFR 10-50 by levels
Gliclazide	50%–100%	20–40 mg/day	20–40 mg/day	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Glipizide	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Guanadrel	q12h	q12-24h	q24-48h	Unknown	Unknown	Dose as GFR 10-50
Guanethidine	q24h	q24h	q24-36h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydralazine	q8h	q8h	q8-12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydroxyurea	100%	50%	20%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydroxyzine	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Idarubicin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ifosfamide	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Iloprost	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

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Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

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Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

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